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Class 11 Chemistry Notes Chapter 3 Classification of elements and periodicity in properties

Introduction:

Classification of elements was proposed in order to study all the elements in a systematic manner. In this Unit, we shall study the development of the Periodic Law and the Periodic Table. Mendeleev's Periodic Table was based on atomic masses. Modern

Periodic Table arranges the elements in the order of their atomic numbers in seven horizontal rows (periods) and eighteen vertical columns (groups or families).

Why Do We Need Classification?

Elements are the basic units of all types of matter. At present, 118 elements are known. With such a large number of elements, it is very difficult to study individually the chemistry of all these elements and their number of compounds. So to make the study of chemistry simpler, scientists searched for a systematic way to organise their knowledge by classifying the elements. Main aim behind this classification was to keep the elements of same properties together, so that by studying one element out of that group, we can have general idea about the properties of all the elements in that group.

Periodic Table

Periodic table may be defined as the tabular arrangement of elements in such a way that the elements having same properties are kept together.

Dobereiner's Triads Law

1st attempt towards the classification of elements was made by Johann W. Dobereiner in 1817. He arranged elements in the groups of three and in such a way that the atomic weight of middle element was equal or nearly equal to the average of atomic weights of other two elements.

	ELEMENTS	SYMBOL	ATOMIC MASS
1	Lithium	Li	6.9
	Sodium	Na	23
	Potassium	K	39
2	Calcium	Ca	40.1
	Strontium	Sr	87.6
	Barium	Ba	137.3
3	Chlorine	Cl	35.5
	Bromine	Br	79.9
	Iodine	I	126.9

Drawback: Only limited triads were arranged in this pattern.

Newland's Law of Octaves

In 1865, John Newland observed that in a series of elements arranged in the increasing order of atomic weights, 1st and 8th elements have same properties. Now, a days, 1st

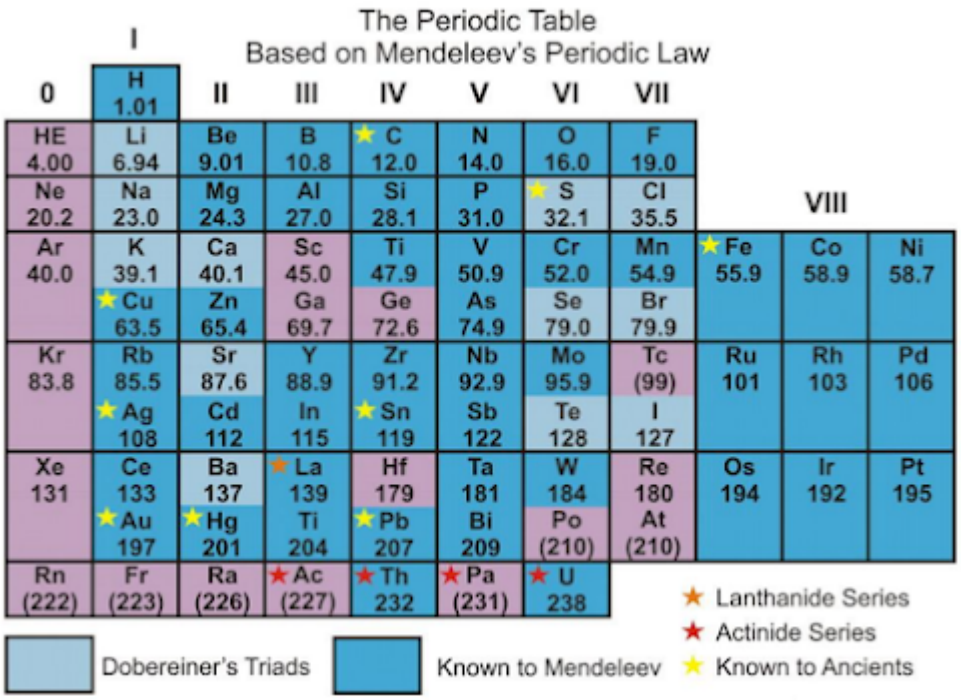
and 9th elements have same properties in that series because noble gases were discovered late.

1	2	3	4	5	6	7	8
Li	Be	B	C	N	O	F	Na
9	10	11	12	13	14	15	16
Mg	Al	Si	P	S	Cl	K	Ca

Mendeleev's Periodic Table

"The physical and chemical properties of elements are a periodic function of atomic weights".

Mendeleev arranged elements in horizontal rows and vertical columns of a table in order of their increasing atomic weights in such a way that the elements with similar properties occupied the same vertical column or group. Vertical Colums are called groups which are numbered I to VIII group, each group is further subdivided into sub groups A and B. Horizontal rows are called **periods**.



Defects in Mendeleev's Table-

i. Position of hydrogen: Position of hydrogen was not justified.

ii. Position of isotope: Isotopes should be placed separately according to periodic law. But actually one place was given to all isotopes of an element.

iii. Cause of periodicity: Mendeleev could not explain why elements exhibit a periodicity in their properties when arranged in the order of increasing atomic weight.

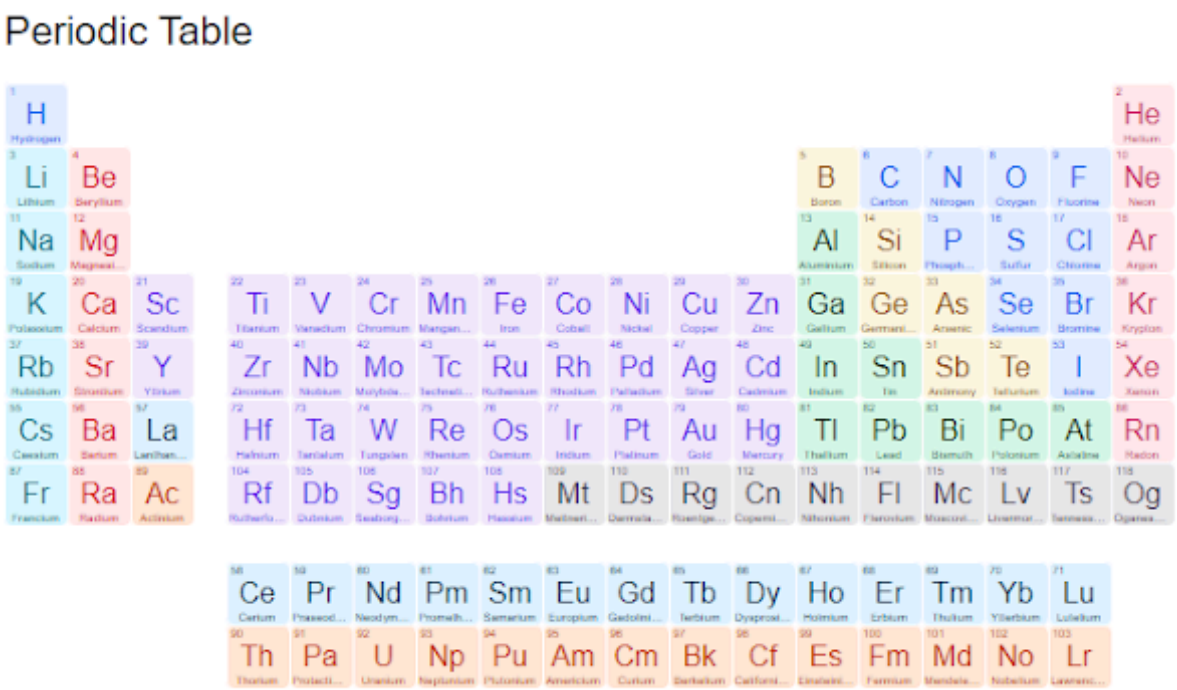
iv. Anomalous pairs of elements: Some anomalous pairs were present in table. As Tellurium (128 u) comes in VI group before Iodine (127 u).

Moseley's

"The physical and chemical properties are the periodic function of their atomic numbers". The long form of periodic table, also called Modem Periodic Table, is based on Modern periodic law. In this table, the elements have been arranged in order of increasing atomic numbers.

A modern version, the so-called **"long form"** of the **Periodic Table** of the elements, is the most convenient and widely used. The horizontal rows are called periods and the vertical columns, groups. Elements having similar outer electronic configurations in their atoms are arranged in vertical columns, referred to as groups or families. According to the recommendation of International Union of Pure and Applied Chemistry (IUPAC), the groups are numbered from 1 to 18 replacing the older notation of groups IA… VIIA, VIII, IB… VIIB and 0.

There are altogether seven periods. The period number corresponds to the highest principal quantum number (n) of the elements in the period. The first period contains 2 elements. The subsequent periods consists of 8, 8, 18, 18 and 32 elements, respectively. The seventh period is incomplete and like the sixth period would have a theoretical maximum of 32 elements. In this form of the Periodic Table, 14 elements of both sixth and seventh periods (lanthanoids and actinoids, respectively) are placed in separate panels at the bottom.



The IUPAC names are derived by using roots for three digit atomic number of the elements.

Digit	IUPAC Name	Symbols
0	nil	n
1	un	u

2	bi	b
3	tri	t
4	quad	q
5	pent	p
6	hex	h
7	sept	s
8	oct	o
9	enn	e

A systematic nomenclature be derived directly from the atomic number of the element using the numerical roots for 0 and numbers 1-9. The roots are put together in order of digits which make up the atomic number and “ium” is added at the end. The IUPAC names for elements with Z above 100 are shown below:

Atomic Number	Name according to IUPAC nomenclature	Symbol	IUPAC Official Name	IUPAC Symbol
101	Unnilunium	Unu	Mendelevium	Md
102	Unnilbium	Unb	Nobelium	No
103	Unniltrium	Unt	Lawrencium	Lr
104	Unnilquadium	Unq	Rutherfordium	Rf
105	Unnilpentium	Unp	Dubnium	Db
106	Unnilhexium	Unh	Seaborgium	Sg
107	Unnilseptium	Uns	Bohrium	Bh
108	Unniloctium	Uno	Hassium	Hs
109	Unnilennium	Une	Meltnerlum	Mt
110	Ununnillium	Uun	Darmstadtium	Ds
111	Unununnium	Uuu	Rontgenium	Rg
112	Ununbium	Uub	Copernicium	Cn
113	Ununtrium	Uut	Nihonium	Nh
114	Ununquadium	Uuq	Flerovium	Fl
115	Ununpentium	Uup	Moscovium	Mc
116	Ununhexium	Uuh	Livermorium	Lv
117	Ununseptium	Uus	Tennessine	Ts
118	Ununoctium	Uuo	Oganesson	Og

Division of Elements into Blocks

s-block: The elements of Group 1 (alkali metals) and Group 2 (alkaline earth metals) which have ns1 and ns2 outermost electronic configuration belong to the s-Block Elements.

Characteristics of s-Block elements,

i. Except Be and Mg, all impart characteristic colour to the flame.
ii. These have low ionisation energy.
iii. These are highly reactive.
iv. These are the highly electropositive elements.
v. All the elements are soft metals.
vi. They have low melting and boiling points.

p-block: The p-Block Elements comprise those belonging to Group 13 to 18 and these together with the s-Block Elements are called the Representative Elements or Main Group Elements. The outermost electronic configuration varies from ns2np1 to ns2np in each period.

Characteristics of p-Block elements,

1. The compounds of p-block elements are generally covalent although their ionic character increases down the group.
2. From left to right 13 to 18, reducing character decreases.
3. The p-block elements generally show more than one oxidation state.
4. The reactivity of elements in a group generally decreases downwards.
5. At the end of each period is a noble gas element with a closed valence shell ns^2 np^6 configuration.

d-block: These are the elements of Group 3 to 12 in the centre of the Periodic Table. These are characterised by the filling of inner d orbitals by electrons and are therefore referred to as d- Block Elements. These elements have the general outer electronic configuration $(n-1)d^{1-10}$ ns^{0-2}

Characteristics of d-Block elements:

1. They are all metals with high melting and boiling points.
2. The compounds of the elements are generally paramagnetic in nature.
3. They mostly form coloured ions, exhibit variable valence (oxidation states).
4. They are of tenly used as catalysts.
5. These elements have high melting point.

f-block: The two rows of elements at the bottom of the Periodic Table, called the **Lanthanoids,** Ce(Z = 58) – Lu(Z = 71) and **Actinoids,** Th(Z=90) – Lr (Z=103) are characterised by the outer electronic configuration $(n-2)f^{1-14}$ $(n-1)d^{0-1}ns^2$. The last electron added to each element is filled in f-orbital. These two series of elements are hence called the **Inner-Transition Elements** (f- Block Elements).

Characteristics of f-Block elements:

1. All actinoids are radioactive. Elements after uranium are known as transuranium element.
2. They form coloured compounds.
3. These two series of elements are called Inner Transition Elements (f-Block Elements).
4. They are all metals. Within each series, the properties of the elements are quite similar.
5. They generally have high melting and boiling points.

Periodic properties.

The properties which generally have a regular trend along a group or period are called periodic properties. These are as given below:

- Atomic size
- Ionisation energy
- Electron gain enthalpy
- Electronegativity

i. Atomic Size: Atomic Radius is the distance from the centre of the nucleus to the outermost shell containing electron.

Ionic Radius: The ionic radii can be estimated by measuring the distances between cations and anions in ionic crystals. In general, the ionic radii of elements exhibit the same trend as the atomic radii.

Cation: The removal of an electron from an atom results in the formation of a cation. The radius of cation is always smaller than that of the atom.

Anion: Gain of an electron leads to an anion. The radius of the anion is always larger than that of the atom.

ii. Ionisation energy: It is the amount of energy required to remove the outer most electron from an isolated atom in its gaseous state. It is the measured in the unit of kJ/mole. It is denoted by (⊿ ¡H).

$M(g) - e^- \rightarrow M^+(g)$

iii. Electron gain enthalpy: It is the enthalpy change when an electron is added to the gaseous neutral atom. Electron gain enthalpy provides a measure of the ease with which an atom adds an electron to form anion. It is the measured in the unit of kJ/mole. It is denoted by ($\Delta_{eg}H$).

$X(g) + e^- \rightarrow X^-(g)$

iv. Electronegativity: Electronegativity is a measure of the tendency of an element to attract bonded electron pair towards itself in a covalently bonded molecule.

Periodic Trends in Chemical Properties along a Period

1. **Metallic character:** It decreases across a period, maximum on the extreme left (alkali metals).
2. **Non-metallic character:** It increases along a period, from left to right.

3. **Atomic Size:** It decreases across a period.
4. **Ionisation energy:** It increases along a period.
5. **Electron gain enthalpy:** It increases along a period.
6. **Electronegativity:** It increases along a period.
7. **Basic nature of oxides:** It decreases from left to right in a period.
8. **Acidic nature of oxides:** It increases from left to right in a period.

Variation in Chemical Properties along a Group

1. **Metallic character:** Generally increases because increase in atomic size and hence decrease in the ionizatiorn energy of the elements in a group from top to bottom.
2. **Non-metallic character:** It generally decreases down a group. As electronegativity of elements decreases from top to bottom in a group.
3. **Atomic Size:** It increases along a group.
4. **Ionisation energy:** It decreases across a period.
5. **Electron gain enthalpy:** It decreases across a period.
6. **Electronegativity:** It decreases across a period.
7. **Basic nature of oxides:** Since metallic character or electropositivity of elements increases in going from top to bottom in a group basic nature of oxidise naturally increases.
8. **Acidic character of oxides:** It generally decreases as non-metallic character of elements decreases in going from top to bottom in a group.
9. **Reactivity of metals:** It generally increases down a group. Since tendency to lose electron increases.

Historical Development Of Periodic Table

Periodic table may be defined as the table which classified all the known elements in accordance with their properties in such a way that elements with similar properties are grouped together in the same vertical column and dissimilar elements are separated from one another.

Doebereiner's Triad

Group	Elements and their Atomic Mass			Arithmetic mean of Atomic mass
A	Lithium(Li)	Sodium(Na)	Potassium(K)	(7.0+39.0)/2= 23.0
	7.0	23.0	39.0	
B	Calcium (Ca)	Strontium(Sr)	Barium(Ba)	(40.0+137.0)/2= 88.5
	40.0	87.5	137.0	
C	Chlorine(Cl)	Bromine (Br)	Iodine(I)	(35.0+127.0)/2= 81.0
	35.0	80.0	127.0	
	55.8	58.9	58.6	

The first attempt towards the classification of elements was made by Johann Wolfgang Doebereiner, a German chemist in 1817.

He arranged similar elements in groups of three and showed that their atomic weights were either nearly the same or the atomic weight of the middle element was approximately the arithmetic mean of the other two. These group of three elements were called Doebereiner's Triads and this generalisation was called Law of triads.

Drawback

The concept of triads could be applied only to a limited number of elements.

Newlands's law of octaves

In1865, an English chemist, John Alexander Newlands's observed at that:

When the lighter elements were arranged in order of their increasing atomic weights, the properties of every 8th elements were similar to those of first one like the eighth note of a musical scale. This is called as Newlands's law of octaves.

Drawback

1) It was applicable to only lighter elements having atomic weights up to 40 u.

2) With the discovery of noble gases, the properties of the 8th element were no longer similar to those of the first one.

Lothar Meyer arrangement

In 1869, Lothar Meyer, a German chemist, studied the physical properties of various element.
He Plotted a graph between the atomic volume and atomic weights of the elements and observed that the elements with similar properties occupied similar position on the curve.
1) The most strongly electropositive alkali metal occupy the peaks on the curve.
2) The less strongly electropositive alkaline earth metals occupy the descending position on the curve.
3) The most electronegative elements i.e. halogens occupy the ascending position on the curve.
Lothar Meyer proposed that the physical properties of the elements are a periodic function of the atomic weights.
He arranged the elements in the tabular form in order of their increasing atomic weights.

Electronic configuration Of Elements

(1) The names are derived directly from the atomic numbers using numerical root for 0 and numbers from 1-9 and adding the suffix ium. The roots for the numbers 0-9 are:

Digit	0	1	2	3	4	5	6	7	8	9
Root	nil	un	bi	Tri	quad	pent	hex	sept	oct	enn
Abbreviation	n	u	b	t	q	p	h	s	o	e

(2) In certain cases, the names are shortened.bi ium and tri ium are shortened to bium and trium and enn nil shortened to ennil.

(3) The symbol of the element is then obtained from the first letters of the roots of numbers which make up the atomic number of the element.
An electron in an atom is characterised by a set of four Quantum numbers(n, l, m and s) and the Principal quantum number (n) defines the main energy level known as the shell.
Location of any element in the periodic table tells us the quantum number(n and l) of the last orbital filled.

Electronic configuration of elements in period

Each period in the periodic table indicates the value of n for the outermost or the valence shell. The total number of elements in each period is twice the number of orbitals available in the energy level that is being filled.

(1) The first period corresponds to the filling of electrons in the first energy shell i.e. (k shell),n=1.Since this energy shell has only 1 orbital i.e. 1s which can accommodate only 2 electrons, therefore, first period has only 2 elements.

(2) The second period corresponds to the filling of electrons in the second energy shell (L shell) i.e. n=2.This shell has 4 orbitals (one 2s and three 2p) which can accommodate 8 electrons, therefore second period contains 8 electron. It starts with Lithium (Z=3) and ends at neon (Z= 10).

(3) The third period corresponds to the filling of electron in the third shells, i.e. n=3.This shell has 9 orbitals (one 3s, three 3p and five 3d) .3d orbital have even higher energy than 4s orbital. Therefore 3d orbitals are filled only after filling of 4s orbital. Third period involves the filling of only 4 orbitals(one 3s and three 3p) and thus contains 8 elements. It starts with sodium(Z=11) and ends at argon (Z= 18).

(4) The Fourth period corresponds to the filling of electrons in the fourth energy level, n=4. It starts with potassium(Z=19) and ends at calcium (Z= 20).

After the filling of 4s orbitals, the filling of five 3d orbitals begins since the energy of 3d orbital is lower than those of 4p orbitals but higher than that of 4s orbital. The filling of 4d and 4f orbital does not occur in this period since their energies are higher than that of even 5s orbital. The filling of the 3d orbital starts from scandium (Z= 21) and ends at Zinc(Z= 30). These 10 elements constitute the 3d transition series.

The filling of 4p orbital begins at gallium(Z=31)and ends at krypton(Z=36) which has the outer electronic configuration as 4s2 3d10 4p6 .In the 4th period, the filling of only 9 orbitals(one 4s, five 3d and three 4p) occurs which can accommodate at the maximum 18 electrons. Therefore 4th period contain 18 electrons from potassium to Krypton.

(5) The fifth period also contains 18 elements since only 9 orbitals (one 5s, five 4d and three 5p) are available for filling with electrons. It begins with rubidium (Z=37) in which one electron enters 5s orbital. After the filling of 5s orbital, the filling of 4d orbital starts at yittrium (Z=39) and ends at cadmium (Z=48).These ten elements constitute 4d transition series. Filling of 5p orbitals starts at indium (Z= 49) and ends at xenon (Z=54).

(6) The sixth period corresponds to the filling of 6th energy level i.e. n=6. Only 16 orbitals (one 5s, five 4d and three 5p) are available for filling with electrons, therefore

6th period contains 32 elements. It begins with caesium (Z=55) in which one electron enters the 6s orbital and ends up with radon (Z=86) in which the filling of 6p orbital is complete. After the filling of 6s Orbital, the next electron enters the 5d orbital and therefore the filling of seven 4f orbitals begins with Cerium (Z=58) and ends up with lutetium (Z=71).These 14 elements constitutes the first inner transition series called lanthanides or lanthanoids.

Filling of 5d orbitals which started at lanthanum continuous from hafnium (Z=72) till it is filled at mercury (Z=80). These 10 elements constitutes the 5d- transition series. After the filling of 5d orbitals, the filling of 6p orbitals starts at thallium (Z=81) and ends at the radon (Z=86).

(7) The seventh period corresponds to filling of 7th energy shells i.e. n=7. It also contain 32 elements corresponds to the filling of 16 orbitals (one 7s, seven 5f, five 6d and three 7p).

After the filling of 7s orbital, the next two electrons enters the 6d orbitals and therefore the filling of seven 5f orbitals begin with proactinium (Z=91) and ends up with lawrencium (Z=103).

Thorium does not have any electron in the 5f orbital, yet get it is considered to be a f block element since its properties resemble more the f block element than the d block elements. These 14 elements from thorium (Z=90) to lawrencium (Z=103) constitute the second (or 5f) inner transition series which is called as actinides are actinoids.

Filling of 5d orbitals which started at actinium (Z=89) continues till it is completed at these Uub (Z=112).These 10 elements constitute the 6d transition series. The filling of 6d, orbital the filling of 7p orbitals begins at Uut (Z= 118) which ends at Uut (Z=118) which belongs to noble gas family.

The first three periods containing 2, 8, 8 elements and are known as short periods while the next three periods containing 18,18, 32 elements are called Long periods

Group wise electronic configuration

The elements in the same group or vertical column have similar valence shell electron electronic configuration i.e. they have the same number of electrons in the outer orbitals and hence have similar properties. Elements of group 1 all have ns1 valence shell electronic configuration. Elements of group 17 all have ns2 np5 valence shell electronic configuration.

Prediction of period, group and block of a given element

The period of an element corresponds to the principal quantum number of the valence shell.

The block of an element corresponds to the type of orbital which receive the last electron.

The group of an element is predicted from the number of electrons in the valence shell or/and penultimate shell as follows:

a) For s block elements, group number is equal to the number of valence electrons.

b) For p block elements, group number is equal to 10+number of electrons in the valence shell.

c) For d block elements, group number is equal to the number of electrons in a (n-1) sub shell + the number of electrons in valence shell.

Question: Write the electronic configuration of the element with atomic number 29. Predict the period, group number and block to which it belongs.

Answer: Z = 29 $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^1$

4 EARNING

Elements receive the last electron in the 3d orbital, therefore, it belongs to d block elements and its group number = No. of electrons in the penultimate shell and valence shell = 10+1=11

The period of the element No. of principal quantum number of the valence shell = 4th

Summary-

1. Periodic table: Arrangement of elements in the increasing order of atomic number such that elements with similar properties fall under same vertical column.

2. Group: A vertical column of elements in the periodic table.

3. Period: A horizontal row of elements in the periodic table.

4. Long form of periodic table has 18 groups and 7 periods. Sixth period is the longest and first period is the smallest.

5. s-Block elements: Elements of groups 1 and 2. Their general valence shell electronic configuration is ns^{1-2}.

6. p-Block elements: Elements of groups 13, 14, 15, 16, 17 and 18. Their general valence shell electronic configuration is ns^2np^{1-6}.

7. d-Block elements: Elements of groups 3, 4, 5, 6, 7, 8, 9, 10, 11 and 12. Also known as transition elements. Their general valence shell electronic configuration is $(n-1)\ d^{1-10}\ ns^{1-2}$. $_{46}Pd$ is exception ($4d^{10}\ 5s^0$).

8. f-Block elements: The two horizontal rows of elements at the bottom of the table. Also known as inner transition elements. Their general valence shell electronic configuration is $(n-2)f^{1-14}\ (n-1)d^{0-1}\ ns^2$.

9. Covalent radius: Half of the internuclear distance between two atoms of the element held by a single covalent bond.

10. Van der Waal's radius: Half of the internuclear distance between two nearest atoms belonging to two adjacent molecules in solid state.

11. Metallic radius: Half of the internuclear distance between two nearest atoms in the metallic lattice.

12. Isoelectronic ions: The ions having same number of electrons but different nuclear charge.

Example: (i) N^{3-}, O^{2-}, F^-, Na^+, Mg^{2+}, Al^{3+}; (ii) p^{3-}, S^{2-}, Cl^-, K^+, Ca^{2+}, Sc^{3+}

13. Among isoelectric ions, greater the nuclear charge smaller is the size.

14. **Ionization enthalpy:** The minimum amount energy required to remove the outermost electron from an isolated gaseous atom of the element.
15. Ionization enthalpy increases along the period and decreases down the group.
16. Be, Mg, N, P and noble gases have exceptionally high values of ionization enthalpies due to their stable electronic configurations.
17. **Electron gain enthalpy:** The enthalpy change taking place when an electron is added to an isolated gaseous atom of the element.
18. Electron gain enthalpy becomes more negative as we move along the period and becomes less negative down the group.
19. Successive electron gain enthalpies are always positive.
20. Helium has the highest value of ionization enthalpy in periodic table.
21. Chlorine has the highest negative electron gain enthalpy in periodic table.
22. **Electronegativity:** It is the tendency of an atom in a molecule to attract towards itself the shared pair of electrons.
23. Fluorine is the most electronegative element whereas Caesium is the least electronegative element in periodic table.
24. Unlike ionisation energy and electron affinity, electronegativity is the property of atom of an element in combined state.
25. Electropositive or metallic character is related to the ionisation energy of the element. The elements having low I.E. are more electropositive or more metallic in character.
26. **Valence** of an element belonging to s or p-block is either equal to the number of valence electrons or eight minus the number of valence electron.
27. The chemical reactivity is maximum at the two extreme ends of the periodic table and is least in the centre.
28. Among alkali metals reactivity increases on descending the group while among halogens the reactivity decreases on descending the group.
29. The basic character of oxides decreases while the acidic character increases on going from left to right in a period.
30. Oxides of metals are generally basic while that of non-metals are acidic in nature.
31. The similarity between a pair of elements in period second and third located diagonally in the periodic table is called the **diagonal relationship**.

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